ThreatMatrix AI — Master Documentation v1.0

Part 3: Module Specifications & UI/UX Design System

> **Part:** 3 of 5 | **Version:** 1.0.0 | **Date:** 2026-02-23
> **Prev:** Part 2 — Architecture | **Next:** Part 4 — ML & LLM

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1. UI/UX Design System

1.1 Design Philosophy: Intelligence Agency Operations Center

Every pixel must reinforce the feeling of being inside a **cyber operations center**. The user should feel like they are commanding a defense system, not browsing a dashboard.

1.2 Color System

```
:root {
  /* Core Palette */
  --bg-primary: #0a0a0f; /* Deep black - operations center base */
  --bg-secondary: #111118; /* Slightly lighter panels */
  --bg-tertiary: #1a1a24; /* Card backgrounds */
  --bg-elevated: #222230; /* Elevated panels, modals */

  /* Accent Colors */
  --cyan: #00f0ff; /* Primary - cyber intelligence */
  --cyan-dim: #00f0ff33; /* Cyan with transparency */
  --cyan-glow: 0 0 20px #00f0ff44, 0 0 40px #00f0ff22;

  /* Status Colors */
  --critical: #ef4444; /* Critical severity */
  --critical-glow: 0 0 15px #ef444444;
  --high: #f97316; /* High severity (orange) */
```

```
  --warning: #f59e0b; /* Warning / medium severity */
  --safe: #22c55e; /* Safe / resolved / normal */
  --info: #3b82f6; /* Informational */

  /* Text */
  --text-primary: #e2e8f0; /* Primary text */
  --text-secondary: #94a3b8; /* Secondary/dimmed text */
  --text-muted: #475569; /* Muted labels */

  /* Borders & Surfaces */
  --border: #ffffff10; /* Subtle borders */
  --border-active: #00f0ff30; /* Active/focused borders */
  --glass: rgba(255, 255, 255, 0.03); /* Glassmorphism fill */
  --glass-border: rgba(255, 255, 255, 0.06);

  /* Gradients */
  --gradient-threat: linear-gradient(135deg, #ef4444, #f97316);
  --gradient-safe: linear-gradient(135deg, #22c55e, #06b6d4);
  --gradient-cyber: linear-gradient(135deg, #00f0ff, #3b82f6);
}
```

1.3 Typography

```
/* Data/Metrics - Monospace */
@import url("https://fonts.googleapis.com/css2?
family=JetBrains+Mono:wght@300;400;500;600;700&display=swap");

/* UI Labels - Sans-serif */
@import url("https://fonts.googleapis.com/css2?
family=Inter:wght@300;400;500;600;700;800&display=swap");

:root {
  --font-data: "JetBrains Mono", monospace;
  --font-ui: "Inter", -apple-system, sans-serif;

  --text-xs: 0.75rem; /* 12px - Ticker text, labels */
  --text-sm: 0.875rem; /* 14px - Secondary info */
  --text-base: 1rem; /* 16px - Body text */
  --text-lg: 1.125rem; /* 18px - Subheadings */
  --text-xl: 1.25rem; /* 20px - Section titles */
  --text-2xl: 1.5rem; /* 24px - Page titles */
  --text-3xl: 2rem; /* 32px - Hero metrics */
  --text-4xl: 2.5rem; /* 40px - War Room main stats */
}
```

1.4 Component Patterns

Glass Panel (Primary Container)

```
.glass-panel {
  background: var(--glass);
  backdrop-filter: blur(12px);
  border: 1px solid var(--glass-border);
  border-radius: 12px;
  padding: 1.5rem;
}
```

Threat Level Indicator

```
.threat-gauge {
  /* DEFCON-style: SAFE → GUARDED → ELEVATED → HIGH → CRITICAL */
  background: conic-gradient(
    var(--safe) 0deg,
    var(--warning) 120deg,
    var(--high) 200deg,
    var(--critical) 280deg,
    var(--critical) 360deg
  );
  border-radius: 50%;
  position: relative;
}
```

Scan Line Animation

```
@keyframes scanline {
  0% {
    transform: translateY(-100%);
  }
  100% {
    transform: translateY(100vh);
  }
}

.scanline-overlay::after {
  content: "";
  position: absolute;
  width: 100%;
  height: 2px;
  background: linear-gradient(90deg, transparent, var(--cyan-dim), transparent);
  animation: scanline 8s linear infinite;
  opacity: 0.15;
}
```

Pulsing Alert Card

```
@keyframes pulse-critical {
  0%,
  100% {
    box-shadow: 0 0 0 0 rgba(239, 68, 68, 0.4);
  }
  50% {
    box-shadow: 0 0 0 8px rgba(239, 68, 68, 0);
  }
}

.alert-card--critical {
  border-left: 3px solid var(--critical);
  animation: pulse-critical 2s ease-in-out infinite;
}
```

1.5 Layout System

```
┌──────────────────────────────────────────────────────────────────────┐
│ TOP BAR (56px) — Logo, Threat Level Badge, Notifications, User      │
├──────────┬──────────────────────────────────────────────────────────┤
│          │                                                          │
│ SIDE     │              MAIN CONTENT AREA                           │
│ NAV      │              (calc(100vh - 56px - 32px))                 │
│ (64px)   │                                                          │
│          │              Scrollable                                  │
│ Icons    │              Grid-based layout per module                │
│ only     │                                                          │
│          │                                                          │
│ Tooltip│                                                          │
│ on       │                                                          │
│ hover    │                                                          │
│          │                                                          │
├──────────┴──────────────────────────────────────────────────────────┤
│ STATUS BAR (32px) — Capture Status, ML Engine, Feeds, Time (UTC)    │
└──────────────────────────────────────────────────────────────────────┘
```

1.6 Navigation Structure

Icon	Module	Route	Keyboard
🎯	War Room	`/war-room`	`Alt+1`
🔍	Threat Hunt	`/hunt`	`Alt+2`
🌓	Intel Hub	`/intel`	`Alt+3`
📡	Network Flow	`/network`	`Alt+4`
🤖	AI Analyst	`/ai-analyst`	`Alt+5`

Icon	Module	Route	Keyboard
🚨	Alert Console	`/alerts`	`Alt+6`
🔬	Forensics Lab	`/forensics`	`Alt+7`
🧠	ML Operations	`/ml-ops`	`Alt+8`
📊	Reports	`/reports`	`Alt+9`
⚙️	Administration	`/admin`	`Alt+0`

---

2. War Room Module

Route: `/war-room` | **Priority:** 🟤 P0 (Crown Jewel) | **Owner:** Full-Stack Dev + Lead Architect

2.1 Purpose

The primary operational view. This is the first page any user sees and the centerpiece of every demo. It must make jaws drop.

2.2 Layout Grid

```
┌──────────────────────────────────────────────────────────────────────────┐
│  THREAT LEVEL: ████ ELEVATED          [🟤 3 Critical] [Auto-refresh]│
├──────────────────────────────────────┬──────────────────────────────────┤
│                                      │  LIVE METRICS (4 cards)          │
│    MAIN THEATER:                     │  ┌─────────┐ ┌─────────┐         │
│    LIVE THREAT MAP                 │  │Pkts/s   │ │Flows    │         │
│                                      │  │12,847   │ │342      │         │
│    Deck.gl ScatterplotLayer          │  └─────────┘ └─────────┘         │
│    + ArcLayer for attack flows       │  ┌─────────┐ ┌─────────┐         │
│    + HeatmapLayer for hotspots       │  │Anomaly% │ │Threats  │         │
│                                      │  │3.2%     │ │48 (24h) │         │
│    Dark Maplibre basemap             │  └─────────┘ └─────────┘         │
│    Auto-center on activity           │                                  │
│                                      │  PROTOCOL DISTRIBUTION           │
│                                      │  (Animated donut chart)          │
│                                      │  TCP ═══════ 62%                 │
│                                      │  UDP ═══     24%                 │
│                                      │  ICMP ═      8%                  │
├──────────────────────────────────────┤  Other ═     6%                  │
│  LIVE ALERT FEED (scrolling)         ├──────────────────────────────────┤
│  🟤 22:14:03 DDoS detected...        │  TRAFFIC TIMELINE (area chart)   │
│  🟡 22:13:58 Suspicious DNS...       │  Glowing cyan area               │
│  🟡 22:13:42 Port scan from...       │  Anomaly spikes in red overlay   │
│  🟤 22:13:30 Baseline restored       │  Last 60 minutes, auto-scroll    │
├──────────────────────────────────────┴──────────────────────────────────┤
│  AI BRIEFING: "In the last hour, your network processed 46K flows.      │
│  3 anomalous patterns detected..."  [View Full Briefing →]              │
├──────────────────────────────────────────────────────────────────────────┤
│  TOP THREATS      │  TOP TALKERS (IPs)     │  GEO DISTRIBUTION          │
```

```
│  1. DDoS (94%)   │  10.0.1.5    42%      │  ET Ethiopia   65%       │
│  2. Scan (87%)   │  10.0.1.12   28%      │  CN China      18%       │
│  3. DNS (72%)    │  10.0.1.8    18%      │  US USA         9%       │
└──────────────────────────────────────────────────────────────────────┘
```

2.3 Components Specification

Component	Data Source	Update Frequency	Technology
Threat Level Gauge	Alert severity aggregation	5 seconds	SVG gauge with CSS animation
Live Threat Map	Flow geo-IP data	Real-time (WebSocket)	Deck.gl + Maplibre
Metric Cards	Flow/alert aggregation API	3 seconds	Recharts SparkLine
Live Alert Feed	WebSocket `alerts:live`	Real-time	Virtual scrolling list
Protocol Distribution	`/flows/protocols`	10 seconds	Recharts PieChart
Traffic Timeline	`/flows/stats?interval=1m`	5 seconds	Recharts AreaChart
AI Briefing Widget	`/llm/briefing` (cached)	5 minutes	Typing effect animation
Top Threats	`/alerts/stats`	10 seconds	Ranked list with bars
Top Talkers	`/flows/top-talkers`	10 seconds	Horizontal bar chart
Geo Distribution	Flow geo-IP aggregation	30 seconds	Flag list with percentages

2.4 Map Layers (Deck.gl)

Layer	Purpose	Visual
ScatterplotLayer	Plot source/destination IPs on map	Cyan dots (normal), Red dots (anomaly)
ArcLayer	Show connection paths between IPs	Glowing arcs, color-coded by severity
HeatmapLayer	Show geographic concentration	Heat gradient overlay
IconLayer	Mark known threat actor locations	Skull icons for known bad IPs

3. Threat Hunt Module

Route: `/hunt` | **Priority:** 🔴 P0 | **Owner:** Lead Architect

3.1 Purpose

Proactive threat investigation tool. Analysts can build queries, filter by IOC/IP/protocol/timeframe, and drill into suspicious patterns.

3.2 Layout

```
┌──────────────────────────────────────────────────────────────────────┐
│  THREAT HUNT                                [Save Query] [History ▾]  │
├──────────────────────────────────────────────────────────────────────┤
│  QUERY BUILDER                                                        │
│  ┌──────────────────────────────────────────────────────────────────┐  │
│  │ Source IP [_________] Dest IP [_________] Protocol [▾ Any]     │  │
│  │ Port Range [___-___] Time Range [Last 1h ▾] Severity [▾ Any]  │  │
│  │ Min Score [0.5____] Label [▾ Any]  [🔍 Hunt]  [Reset]        │  │
│  └──────────────────────────────────────────────────────────────────┘  │
├──────────────────────────────────────────────────────────────────────┤
│  RESULTS (2,847 flows matching)        [Export CSV] [Analyze w/ AI] │
│  ┌──────────────────────────────────────────────────────────────────┐│
│  │ Time       │ Src IP      │ Dst IP      │ Proto│ Score │ Label  ||
│  │ 22:14:03  │ 10.0.1.5    │ 8.8.8.8     │ UDP  │ 0.94  │ DDoS   ||
│  │ 22:13:58  │ 10.0.1.5    │ evil.com    │ TCP  │ 0.87  │ C2     ||
│  │ 22:13:42  │ 192.168.1.1 │ 10.0.1.100  │ TCP  │ 0.72  │ Scan   ||
│  └──────────────────────────────────────────────────────────────────┘│
├──────────────────────────────────────────────────────────────────────┤
│  FLOW DETAIL (selected)                                               │
│  Full feature vector │ Related alerts │ Intel matches │ Timeline    │
└──────────────────────────────────────────────────────────────────────┘
```

3.3 Key Features

Feature	Description
Query Builder	Visual filter builder with AND/OR logic, saved queries
Results Table	Sortable, paginated, with inline sparklines for traffic patterns
Flow Detail Drawer	Side panel showing all 40+ features, related alerts, IOC matches
AI Analysis Button	Send selected flows to AI Analyst for natural language analysis
Export	CSV/JSON export for offline analysis
Hunt History	Saved previous queries with timestamps
Quick Filters	Pre-built: "Unusual DNS", "Port Scans", "High Volume", "Foreign IPs"

---

4. Intel Hub Module

Route: `/intel` | **Priority:** 🟤 P0 | **Owner:** Lead Architect

4.1 Purpose

Threat intelligence aggregation and IOC (Indicators of Compromise) management. Correlates external intelligence with internal network observations.

4.2 Components

Component	Description
IOC Browser	Searchable/filterable table of all IOCs from feeds (IP, domain, hash)
IP Reputation Lookup	Enter any IP → get reputation score, associated threats, feed sources
Feed Status Dashboard	Health of each feed (OTX, AbuseIPDB, VirusTotal), last sync, IOC count
Correlation Panel	IOCs matched to internal network flows — "These IOCs were seen in your traffic"
Threat Actor Profiles	Known threat groups associated with detected IOCs
Manual IOC Entry	Add custom IOCs (e.g., from incident response)

4.3 Feed Integration Details

Feed	API	Free Tier	Data Retrieved
AlienVault OTX	REST	Unlimited (with key)	Pulses, IP/domain/URL IOCs, threat descriptions
AbuseIPDB	REST	1,000 checks/day	IP reputation, abuse reports, categories
VirusTotal	REST	500 requests/day	IP/domain/file analysis, detection ratios

5. Network Flow Module

Route: `/network` | **Priority:** P0 | **Owner:** Full-Stack Dev

5.1 Components

Component	Description	Chart Type
Traffic Volume Timeline	Packets/bytes over time with anomaly overlay	Area chart (Recharts)
Top Talkers	Top source/destination IPs by traffic volume	Horizontal bar chart
Protocol Distribution	Breakdown by protocol (TCP/UDP/ICMP/Other)	Pie/donut chart
Port Activity	Most active destination ports	Treemap
Connection Graph	Force-directed graph of IP connections	D3 force layout or Deck.gl
Bandwidth Monitor	Real-time throughput (Mbps in/out)	Dual area chart
Geo Traffic Map	Traffic distribution by country	Choropleth map
Flow Table	Live-updating sortable table of recent flows	Virtual scrolling table

6. AI Analyst Module

Route: `/ai-analyst` | **Priority:** P0 | **Owner:** Lead Architect

6.1 Purpose

The **innovation differentiator**. A conversational AI interface that lets analysts investigate threats using natural language. No other cybersecurity senior project has this.

6.2 Layout

```
┌──────────────────────────────────────────────────────────────────────────┐
│  🤖 AI ANALYST          [Context: General ▾]  [Token Budget: 82%] │
├──────────────────────────────────┬──────────────────────────────────────┤
│                                  │  QUICK ACTIONS                       │
│  CONVERSATION                    │  ┌──────────────────────────┐        │
│                                  │  │ 📋 Daily Threat Briefing  │        │
│  ┌──────────────────────────┐    │  │ 🔍 Analyze Latest Alert   │        │
│  │ 🤖 AI: Based on analysis  │    │  │ 📊 Network Health Summary │        │
│  │ of the last 2 hours, I've │    │  │ ET Translate Last Report  │        │
│  │ identified 3 anomalous    │    │  │ 🎯 Top Risks Assessment   │        │
│  │ patterns in your traffic: │    │  │ 📝 Generate Exec Summary  │        │
│  │                           │    │  └──────────────────────────┘        │
│  │ 1. DNS tunneling suspect  │    │                                      │
│  │    from 10.0.1.23 (conf:  │    │  CONTEXT PANEL                       │
│  │    87%)                   │    │  Shows data AI is referencing:       │
│  │ 2. Port scan activity...  │    │  • Related flows                     │
│  │                           │    │  • Alert details                     │
│  │ 👤 You: What about the    │    │  • IOC matches                       │
│  │ DNS traffic specifically? │    │  • Model predictions                 │
│  │                           │    │                                      │
│  │ 🤖 AI: The DNS queries    │    │  CONFIDENCE INDICATOR                │
│  │ from 10.0.1.23 show...    │    │  ▓▓▓
```

Query Type	Example Prompt	Backend Action
Report Generation	"Generate a daily briefing"	Aggregate 24h data, LLM writes report
Translation	"Translate the last alert to Amharic"	Fetch alert text, send to LLM for translation
Playbook	"How do I respond to this DDoS?"	LLM generates incident response steps

6.4 Streaming Response

LLM responses stream token-by-token via WebSocket + Server-Sent Events for a typing effect that makes the AI feel alive and responsive.

---

7. Alert Console Module

Route: `/alerts` | **Priority:** P1 | **Owner:** Full-Stack Dev

7.1 Alert Lifecycle

```
New Alert → OPEN → ACKNOWLEDGED → INVESTIGATING → RESOLVED
                                                  → FALSE POSITIVE
```

7.2 Components

Component	Description
Alert Table	Sortable by severity/time/status, with search and filters
Alert Detail Drawer	Full alert info, AI narrative, related flows, timeline
Severity Filter Tabs	Critical / High / Medium / Low / Info / All
Status Filter	Open / Acknowledged / Investigating / Resolved
Bulk Actions	Acknowledge multiple, assign to analyst
Alert Timeline	Chronological view of alert activity
Statistics Panel	Alert counts by severity, MTTR, trends

7.3 Severity Definitions

Level	Color	Trigger	Response Time SLA
Critical	`#ef4444`	Active attack detected (DDoS, C2), confidence ≥90%	Immediate
High	`#f97316`	Likely attack (port scan, brute force), confidence ≥75%	15 minutes

Level	Color	Trigger	Response Time SLA
Medium	`#f59e0b`	Suspicious activity, confidence ≥50%	1 hour
Low	`#3b82f6`	Unusual but potentially benign pattern	4 hours
Info	`#94a3b8`	Informational observation	Review at convenience

8. Forensics Lab Module

Route: `/forensics` | **Priority:** ⚪ P1 | **Owner:** Lead Architect

8.1 Components

Component	Description
PCAP Upload	Drag-and-drop .pcap/.pcapng file upload with progress
Analysis Results	Summary: packet count, flow count, anomalies found, timeline
Packet Inspector	Table of extracted packets with protocol breakdown
Flow Reconstruction	Reconstructed TCP sessions for suspicious connections
Timeline View	Chronological event timeline from PCAP data
AI Analysis	Send PCAP results to AI Analyst for narrative
Upload History	Previous uploads with status and results

9. ML Operations Module

Route: `/ml-ops` | **Priority:** ⚪ P1 | **Owner:** Lead Architect

9.1 Components

Component	Description
Model Registry	Table of all trained models with status (active/retired)
Performance Dashboard	Accuracy, precision, recall, F1 per model
Confusion Matrix	Interactive heatmap for each model's classification matrix
ROC Curves	Overlay ROC curves for model comparison (academic money shot)
Feature Importance	Ranked bar chart of most important features per model
Model Comparison Table	Side-by-side metrics: 3 models × N metrics
Inference Monitor	Real-time inference throughput and latency
Retraining Controls	Trigger retraining with dataset/hyperparameter selection
Training History	Log of all training runs with metrics over time

9.2 The Academic Money Shot: Model Comparison

```
┌──────────────────────────────────────────────────────────────────────────┐
│  MODEL COMPARISON                                                        │
├──────────────┬──────────────────┬──────────────────┬──────────────────────┤
│ Metric       │ Isolation        │ Random Forest    │ Autoencoder          │
│              │ Forest           │                  │                      │
├──────────────┼──────────────────┼──────────────────┼──────────────────────┤
│ Accuracy     │ 92.3%            │ 97.1%            │ 94.8%                │
│ Precision    │ 89.7%            │ 96.5%            │ 93.2%                │
│ Recall       │ 91.1%            │ 97.8%            │ 95.1%                │
│ F1-Score     │ 90.4%            │ 97.1%            │ 94.1%                │
│ AUC-ROC      │ 0.945            │ 0.991            │ 0.968                │
│ Train Time   │ 2.3s             │ 8.7s             │ 45.2s                │
│ Inference/ms │ 0.8ms            │ 1.2ms            │ 3.5ms                │
│ Type         │ Unsupervised     │ Supervised       │ Deep Learning        │
│ Zero-Day     │ ☑ Yes            │ ✕ No             │ ☑ Yes                │
│ Explainable  │ Partial          │ ☑ Full           │ ✕ Limited            │
├──────────────┴──────────────────┴──────────────────┴──────────────────────┤
│ Champion Model: Random Forest v2 (highest F1 on CICIDS2017)              │
│ Anomaly Sentinel: Isolation Forest (best zero-day capability)            │
└──────────────────────────────────────────────────────────────────────────┘
```

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10. Reports Module

Route: `/reports` | **Priority:** 🟢 P1 | **Owner:** Full-Stack Dev

10.1 Report Types

Report	Trigger	Content	Format
Daily Threat Summary	Scheduled (6 AM) or on-demand	24h alerts, top threats, anomaly trends, AI narrative	PDF
Incident Report	Per-alert action	Alert details, timeline, affected IPs, response actions	PDF
Executive Briefing	On-demand	High-level threat posture, risk score, recommendations	PDF
ML Performance Report	On-demand	Model comparison, accuracy trends, retraining recommendations	PDF
Network Health Report	Weekly scheduled	Traffic patterns, protocol distribution, bandwidth trends	PDF
Compliance Report	On-demand	Alert response times, resolution rates, SLA adherence	PDF

---

11. Administration Module

Route: `/admin` | **Priority:** 🟤 P2 | **Owner:** Lead Architect

11.1 Sub-Pages

Sub-Page	Route	Components
User Management	`/admin/users`	User CRUD, role assignment, activity log
System Config	`/admin/config`	Capture settings, alert thresholds, retention policy
LLM Budget	`/admin/llm-budget`	Token usage charts, provider breakdown, budget alerts
Feed Management	`/admin/feeds`	Enable/disable feeds, sync schedules, API key management
Capture Interfaces	`/admin/capture`	Select NICs, BPF filters, capture mode toggle
Audit Log	`/admin/audit`	Full system audit trail (who did what and when)

11.2 LLM Budget Dashboard

```
┌──────────────────────────────────────────────────────────────────────┐
│  LLM BUDGET MONITOR                                                  │
├──────────────────────────────────────────────────────────────────────┤
│                                                                      │
│  TOTAL BUDGET: $150.00    SPENT: $42.38 (28.3%)    REMAINING: $107.62│
│  ████████▒▒▒
```

Module	Status	Deliverable
Network Flow	P0	Traffic analysis panels
Alert Console	P0	Alert listing and detail
AI Analyst	P0	Chat interface with streaming
Intel Hub	P0	IOC browser and lookup
Threat Hunt	P0	Query builder and results

MVP Iteration 2 (Week 5-6): Intelligence Layer

Module	Status	Deliverable
Forensics Lab	P1	PCAP upload and analysis
ML Operations	P1	Model dashboards, comparison
Reports	P1	PDF generation

MVP Iteration 3 (Week 7-8): Polish & Enterprise

Module	Status	Deliverable
Administration	P2	User mgmt, config, budget
i18n	P2	Full Amharic/English toggle
Polish	P2	Animations, responsive, testing

> **End of Part 3** — Continue to Part 4: ML Pipeline, LLM Integration & Data Strategy

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ThreatMatrix AI Master Documentation v1.0 — Part 3 of 5

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